

VARUNA06: An Intelligent Low-Cost Multi-Sensor Platform for Seafloor Mineral Prospectivity Mapping

1. Background and Motivation

Deep-ocean environments host strategically important mineral resources, including polymetallic nodules, seafloor massive sulfides (SMS), cobalt-rich ferromanganese crusts, **and rare-earth-element-bearing sediments**. These resources contain critical metals such as Mn, Ni, Co, Cu, Zn, and rare-earth elements that are increasingly relevant to energy, electronics, and advanced manufacturing technologies. However, conventional deep-sea mineral exploration relies on research vessels, sophisticated remotely operated vehicles (ROVs), autonomous underwater vehicles (AUVs), geophysical instrumentation, and subsequent geological sampling, leading to high survey costs and limited spatial coverage.

A key technological requirement is therefore a **low-cost reconnaissance system capable of identifying and prioritizing mineral-prospective seabed locations before expensive detailed investigation or physical sampling**.

Marine electromagnetic methods are particularly relevant because electrical conductivity contrasts associated with mineralization can produce measurable electromagnetic responses. Gehrmann et al. (2019), for example, demonstrated controlled-source electromagnetic (CSEM) investigation of seafloor massive sulfides at the TAG hydrothermal field and showed that CSEM observations can constrain conductive mineralized structures. More recent studies have demonstrated that integration of transient electromagnetic and self-potential observations can further improve delineation of SMS deposits (Su et al., 2022). Magnetic measurements provide an independent geophysical indicator because hydrothermal alteration modifies the magnetic properties of oceanic crust, producing detectable near-bottom magnetic anomalies.

For polymetallic nodules and crusts exposed at the seabed, acoustic backscatter, bathymetry, and optical imagery provide complementary information regarding seabed morphology, texture, and nodule coverage. Recent studies demonstrate that multisource geological information and machine-learning approaches can improve the quantitative prediction of polymetallic nodule occurrence and abundance.

VARUNA06 is therefore proposed as a low-cost, tethered, modular underwater robotic platform that integrates multiple complementary sensing modalities with intelligent data fusion, adaptive rescanning, and spatial prospectivity mapping. Rather than relying on a single sensor response, the system will combine electromagnetic, magnetic, electrical, acoustic, optical, and environmental measurements to produce an evidence-based **Mineral Prospectivity Index (MPI)** and corresponding confidence level.

2. Aim and Objectives

The primary aim is to **design, develop, and experimentally validate a low-cost deployable underwater system for rapid reconnaissance and mapping of mineral-prospective seabed zones using multi-physics sensing and intelligent sensor fusion**.

The specific objectives are:

1. **To design and develop a modular tethered underwater platform** integrating electromagnetic, magnetic, electrical, acoustic, optical, and environmental sensors for near-seabed mineral exploration.
2. **To develop a multi-sensor data acquisition and processing framework** incorporating sensor calibration, noise filtering, environmental/motion correction, feature extraction, and intelligent sensor fusion for reliable anomaly detection.
3. **To develop an adaptive mineral-prospectivity mapping framework** based on a Mineral Prospectivity Index (MPI) and the **Detect → Score → Decide → Rescan → Confirm → Map** strategy for identifying and prioritizing potential mineralized zones.
4. **To experimentally validate the developed system** through laboratory, controlled-water, and shallow-water trials and evaluate its capability to generate geo-referenced prospectivity maps for targeted follow-up investigation.

3. Proposed Technical Methodology

3.1 Multi-Physics Sensor Architecture

VARUNA06 will employ complementary sensors because no single low-cost sensor can unambiguously discriminate the different classes of seabed mineral deposits.

The primary sensing module will use a **multi-frequency electromagnetic transmitter–receiver (EM Tx/Rx) coil system**. An alternating electromagnetic field generated by the transmitter interacts with electrically conductive material within or immediately beneath the seabed. Changes in received amplitude, phase, and frequency-dependent response will provide features related to the electrical properties of the target and surrounding sediments. Marine CSEM and transient-EM methods have already demonstrated sensitivity to electrically conductive SMS mineralization (Gehrmann et al., 2019; Peng et al., 2020; Su et al., 2022).

A three-axis **magnetometer** will simultaneously measure local magnetic-field perturbations. Magnetic data provide a complementary constraint because hydrothermal alteration and mineralization can produce characteristic magnetic anomalies. The combination of magnetic and EM measurements is particularly attractive because the two measurements respond to different physical properties.

An electrical/conductivity sensing module will provide additional information on local electrical behavior, while acoustic sensing will characterize seabed geometry, roughness, and backscatter. A downward-looking camera with controlled illumination will provide optical confirmation of exposed nodules, crusts, and geological textures. Acoustic and optical observations are especially important for polymetallic nodules, for which recent work has demonstrated quantitative estimation of nodule coverage using acoustic and multisource data.

Environmental and navigation measurements, including pressure/depth, temperature, and IMU-based attitude and acceleration, will be recorded synchronously. These measurements are not merely auxiliary variables; they will be incorporated into the signal correction to prevent

changes caused by vehicle motion, sensor altitude, or environmental conditions from being misinterpreted as mineral anomalies.

3.2 Signal Processing and Feature Extraction

For every survey location (x,y), the raw sensor vector will be represented as

$$\mathbf{X}(x,y) = [\mathbf{EM} (f_1 \dots f_n), \mathbf{B}_x, \mathbf{B}_y, \mathbf{B}_z, \sigma, \mathbf{A}_{\text{acoustic}}, \mathbf{I}_{\text{optical}}, d, T, \theta, a]$$

where EM represents frequency-dependent electromagnetic responses, $\mathbf{B}$ represents magnetic-field components, σ represents electrical conductivity, ($\mathbf{A}_{\text{acoustic}}$) represents acoustic features, ($\mathbf{I}_{\text{optical}}$) represents image-derived features, and (d,T, θ ,a) describe depth, temperature, attitude, and motion, respectively.

Raw measurements will undergo:

sensor calibration → baseline correction → filtering → motion/environmental compensation → feature extraction → normalization → spatial registration.

Frequency-domain features such as amplitude ratio, phase shift, and frequency-response slope will be extracted from EM signals. Magnetic residuals will be calculated after baseline/background removal. Acoustic texture and backscatter features will characterize seabed morphology, while computer-vision algorithms will extract optical texture, shape, and exposed-object features.

This processing stage is essential because seawater conductivity, sensor–seabed separation, platform attitude, and seabed topography can strongly influence marine electromagnetic measurements (Peng et al., 2020).

3.3 Multi-Sensor Fusion and Mineral Prospectivity Index

A major innovation of VARUNA06 is that a mineral anomaly will **not be declared from a single sensor threshold**.

Instead, normalized features from independent sensing modalities will be combined to obtain a Mineral Prospectivity Index:

$$\mathbf{MPI}(x,y) = \sum w_i S_i(x,y)$$

where (S_i) represents the normalized evidence provided by each sensing modality and (w_i) represents experimentally calibrated sensor reliability/importance.

The initial implementation will employ interpretable weighted evidence fusion combined with unsupervised anomaly detection, such as **Isolation Forest**. As labeled experimental data increase, supervised classifiers such as Random Forest, Support Vector Machine, or gradient-boosting methods can be evaluated using cross-validation.

Importantly, the system will distinguish between:

Anomaly detection: “Does this location differ significantly from its geological background?”

and

Prospectivity classification: “Is the combined response consistent with the calibrated response of a particular mineralization class?”

This distinction prevents overclaiming direct elemental identification from indirect geophysical observations.

The final output will consist of both an **MPI (0–100)** and a **confidence score**, determined from sensor agreement, signal-to-noise ratio, repeatability, and spatial persistence.

3.4 Closed-Loop Adaptive Survey

Conventional reconnaissance surveys often follow predetermined tracks irrespective of the information collected. VARUNA06 will instead implement a closed-loop strategy:

SENSE → DETECT → SCORE → DECIDE → MOVE → RESCAN → CONFIRM → MAP

When the initial survey identifies a statistically significant anomaly, the vehicle will be repositioned to collect measurements at multiple orientations and/or frequencies around the candidate target. An anomaly will receive high confidence only when it exhibits:

- (i) agreement between complementary sensors,
- (ii) repeated response during rescanning,
- (iii) spatial persistence across neighboring measurements, and
- (iv) consistency after motion/depth/environmental correction.

This strategy is expected to substantially reduce false positives caused by isolated noise, vehicle motion, or heterogeneous seabed conditions.

4. Prototype Development and Experimental Validation

Development will proceed through four Technology Readiness stages:

Stage I – Sensor Characterization: Individual EM, magnetic, conductivity, acoustic, and optical modules will be calibrated in air and under controlled water conditions using known metallic/non-metallic targets.

Stage II – Integrated Prototype: Sensors will be integrated into a waterproof, modular cartridge with microcontroller-based embedded computing, synchronized acquisition, tethered communication, and basic underwater mobility.

Stage III – Controlled Validation: Blind experiments will be conducted using different target materials, target sizes, burial depths, orientations, sensor heights, and environmental conditions. Repeated measurements will quantify repeatability, sensitivity, and false-alarm behavior.

Stage IV – Shallow-Water Demonstration: The complete platform will perform grid-based and adaptive surveys in a realistic aquatic environment. Results will be converted into spatial MPI maps and compared against known target locations.

Performance will be evaluated quantitatively using **probability of detection, false-positive rate, precision, recall/F1-score, localization error, repeatability, signal-to-noise ratio, and mapping accuracy**. Ablation experiments will also compare single-sensor performance against fused multi-sensor performance to demonstrate the actual benefit of sensor fusion.

5. Innovation, Expected Outcomes, and Impact

The principal innovation of VARUNA06 is **not simply the combination of several sensors on an ROV**, but the integration of sensing, decision-making, and adaptive investigation into a single low-cost reconnaissance architecture.

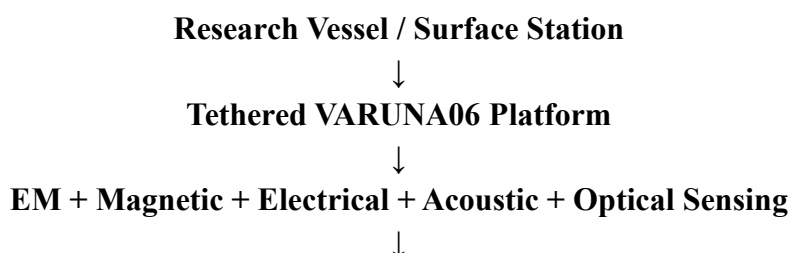
The expected deliverables are:

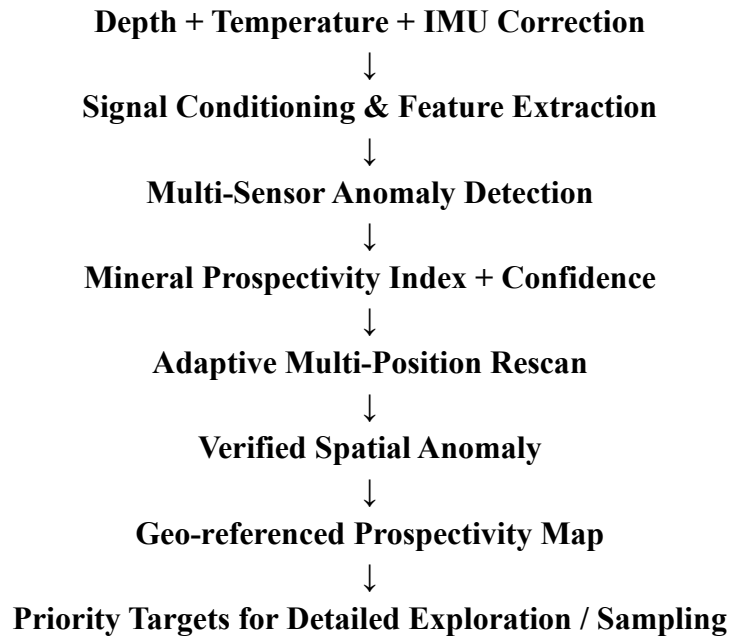
1. A functional modular tethered underwater robotic prototype.
2. Multi-frequency EM and magnetic anomaly-detection modules.
3. Synchronized multi-physics seabed dataset.
4. Motion- and environment-corrected signal-processing pipeline.
5. Multi-sensor Mineral Prospectivity Index with confidence estimation.
6. Autonomous/semi-autonomous anomaly-triggered rescanning logic.
7. Real-time surface dashboard displaying sensor responses and anomaly confidence.
8. Position-linked 2-D seabed prospectivity/anomaly maps.
9. Experimentally quantified detection limits and false-positive performance.
10. A scalable architecture suitable for future pressure-rated deep-ocean deployment.

The system is intended as a **reconnaissance and decision-support platform rather than a replacement for laboratory geochemical analysis or research-grade geophysical surveys**. Its value lies in rapidly screening seabed areas and identifying high-priority targets where expensive detailed surveys, coring or physical sampling should subsequently be concentrated.

This approach can reduce survey time, vessel utilization, and unnecessary seabed intervention while enabling research institutions and marine agencies to conduct more systematic preliminary exploration. The modular architecture further allows future integration of higher-grade sonar, induced-polarization/self-potential sensors, Doppler velocity logging, improved navigation, and deep-water pressure-rated electronics.

Proposed Technology Workflow





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